Battery Management System For EV System

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Project Objective

Develop a safe and functional lithium-ion battery pack with an embedded monitoring system. The EV system built around it serves only as a practical load to validate battery performance.

- Design & assembly of a 7S3P 25.2V battery pack using Samsung 21700 cells.
- Integration of a smart BMS for protection and telemetry.
- Real-world discharge validation using a BLDC hub motor + RMCS-6611 motor driver.
- Telemetry via ESP32-S3 / UNO R4 WiFi with OLED and cloud dashboard.
- Mechanical design and fabrication of the vehicle frame inspired by Fusion 360 model.

Cell Selection

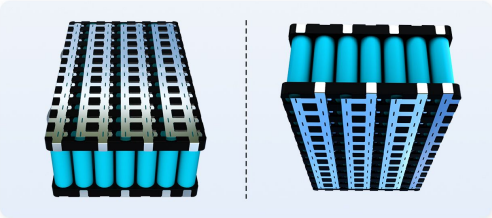
Samsung INR21700-50S (3.6V, 5000mAh, 9C)

- Chosen for high capacity and stable discharge profile.
- Suitable for small EV-scale load testing.
- Nominal: 3.6V, Max: 4.2V, 5000mAh, 9C discharge rating.

Battery Pack Configuration (7S3P)

Practical Configuration Used

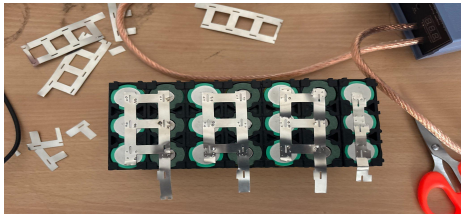
- 7 cells in series → EV-compatible voltage (~25.2V).
- 3 cells in parallel → capacity increased to 15Ah.
- Total pack energy ≈ 378Wh.
- Matches the voltage/current requirements of the BLDC hub motor.

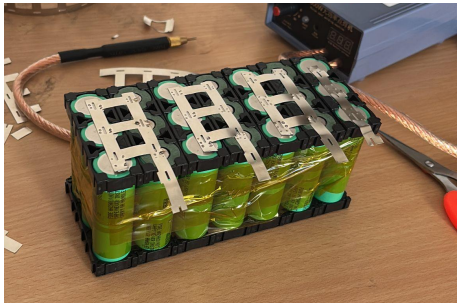


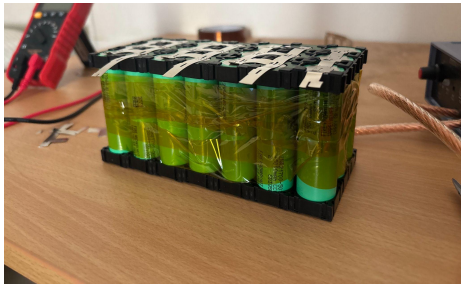
Battery Pack Assembly

Physical Construction Steps

- Cells arranged in **21700 holders** for spacing & safety.
- Spot-welded using **0.15 × 27 mm nickel strip** for series & parallel joints.
- Insulated and reinforced using **Kapton polyimide tape**.
- Focus on safe assembly, mechanical stability, and clean bus-bar routing.



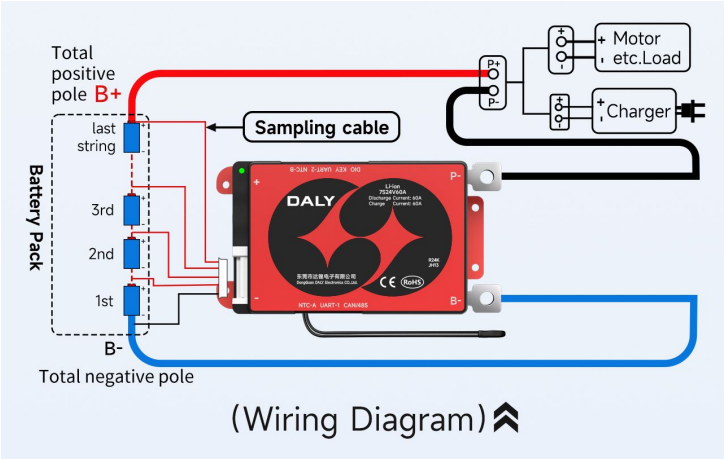




Embedded Battery Monitor Integration

(7S, 30A)

- Overcharge / over-discharge protection
- Overcurrent & short-circuit protection
- Passive cell balancing
- UART telemetry output verified on test bench



(Wiring Diagram)

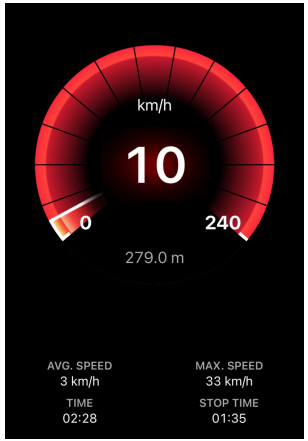
Charging Setup

29.4V 5A Lithium-Ion Charger

- Standard CC–CV charger compatible with 7S configuration.
- Connected via XT60 for safety and low resistance.
- Observed stable charging curve during testing.

EV Load System (Hub Motor + Driver)

Although the project focuses on battery development, a complete EV system was built to apply real mechanical load.



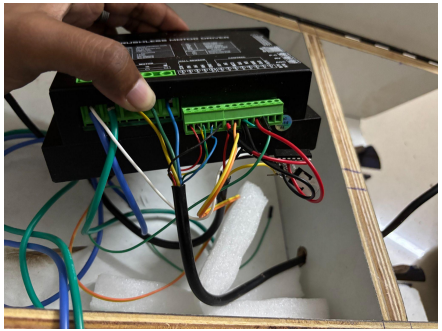
Hub Motor Performance

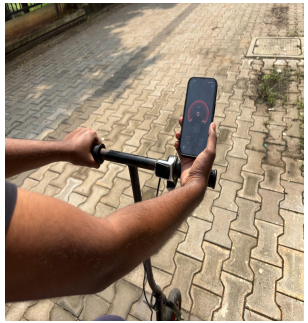
- Rated: 24V, 350W
- **No-load speed:** 10–12 km/h
- **On-load speed:** 5–7 km/h (limited by motor power, battery capability)

RMCS-6611 Motor Driver (750W, 24–48V)

Used as the traction interface:

- Supports hall-sensor input
- Provides throttle input control
- High-current output validation for battery discharge
- Included features: speed control, protection logic, soft start





Dual Control Modes

The EV system can be controlled in **two independent ways**:

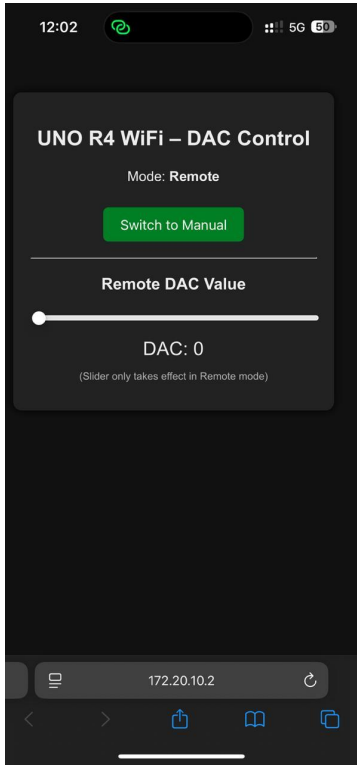
1 Throttle Mode

- Standard EV twist throttle connected to RMCS-6611.
- Provides analog speed control for load testing.

2 PWM Control via UNO (Local IP Webpage)

- UNO R4 WiFi generates PWM signals to the driver.
- A local IP interface allows speed control through a browser.
- Useful for automated, repeatable load patterns.

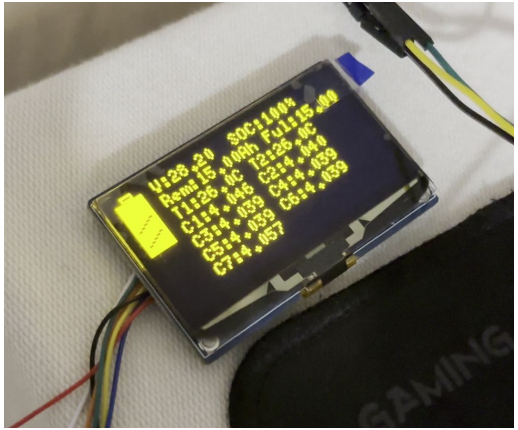
This flexibility allowed controlled and repeatable discharge profiles for battery evaluation.



Telemetry & Microcontroller Integration

UNO R4 WIFI – Practical DAQ Platform

- Connected via UART to BMS to read:
  - Cell voltages
  - Pack voltage, current
  - SOC/SOH
  - Protection status
- Displayed live data on **Waveshare 2.42" OLED**.



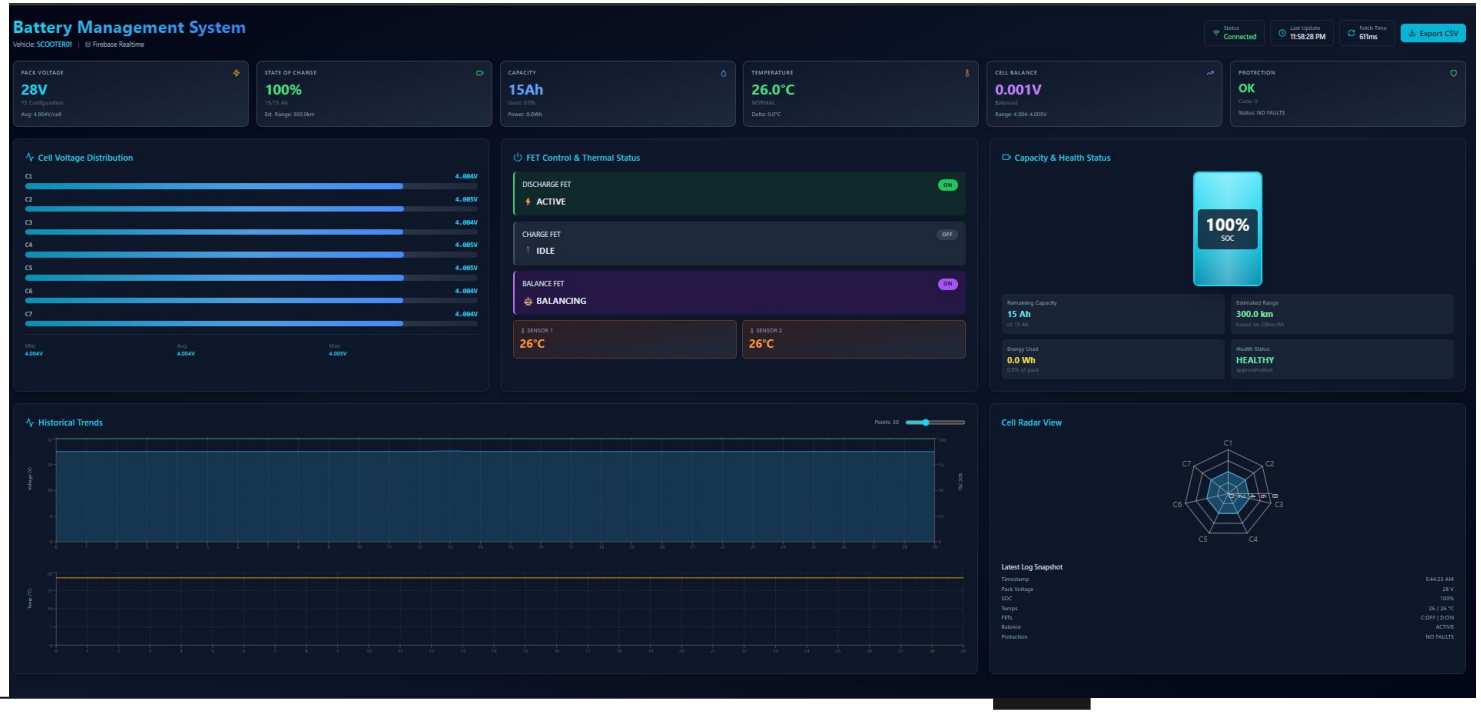
Mechanical Fabrication

- Vehicle frame fabricated to match the layout designed in **Autodesk Fusion 360**.
- Battery, controller, and wiring compartments modeled digitally before fabrication.
- Achieved close alignment with the conceptual design, ensuring stable mounting and safe testing conditions.

Dashboard Development

ReactJS + Firestore

- Cloud database (Firebase) used to log battery & motor parameters.

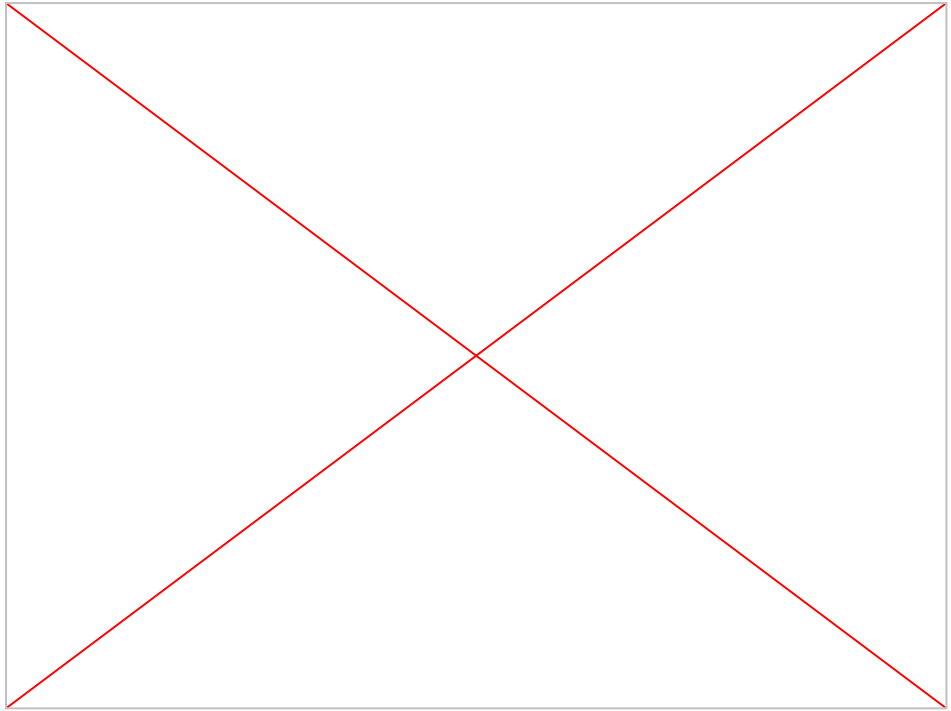


System Workflow

Realistic End-to-End Flow (Built + Tested)

- **Battery Pack (7S3P)**
  → **BMS**
  → **UNO R4 WIFI**
  → **OLED Display**
  → **BLDC Driver + Hub Motor Load**
  → **FireBase Cloud**
  → **React Dashboard**

VIDEO



Thank You